MILITARY SPECIFICATION

BARS, COPPER, SILVER-BEARING (FOR COMMUTATORS)

This specification has been approved by the Naval Ordnance Systems Command, Department of the Navy

1. SCOPE

1.1 This specification describes the requirements for the procurement of silver-bearing copper bars for fabrication into commutator bars. (See 6.1.)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

STANDARDS

Federal

*FED-STD-151	Metals; Test Methods

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

(Copies of specifications, standards, drawings, and publications, required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications The following documents form a part of this specification to the extent specified herein Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards

*E 18	Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials, Tests for
*B 193	Resistivity of Electrical Conductor Materials, Test for

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, Pa. 19103.)

3. REQUIREMENTS

*3.1 Manufacture. Bars shall be manufactured by hot working and shall be finished by such cold rolling or cold drawing, including straightening, as may be required to produce material conforming to the requirements of this specification.

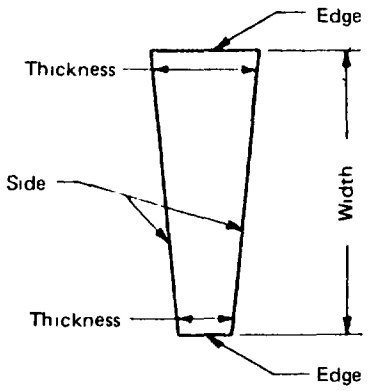
3.2 Chemical composition.

*3.2.1 Copper. The bars shall contain not less than 99.90 percent copper (including silver).

3.2.2 Silver. The bars shall contain not less than 0.085 percent silver (25 Troy ounces per ton).

3.3 Shape. The cross section of the bars shall be that of a wedge, or segmental section, having planar sides and edges. The edges shall be substantially parallel to each other and perpendicular to the plane passing midway between the two sides. A representative section defining the surfaces is shown below. Dimensions shall be as specified on the applicable drawings, contract, or order.

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*3.4 Resistivity. Resistivity at 20° centigrade (C) (68° Fahrenheit (F)) shall not exceed 875.2 ohm-pounds per square mile (conductivity 100% IACS minimum at 20° C).

3.5 Hardness The hardness at any point on the surface of the bars shall be not less than the amount shown in table I.

*Table I

ROCKWELL B HARDNESS

Thickness, thick edge (inch)	Hardness (minimum)
Up to 0.250 incl.	35
Over 0.250	27

3.6 Dimensional variations.

3.6.1 Thickness. The thickness at points 0.250 inch from either edge shall not vary from the specified thickness by more than ±0.001 inch, regardless of width.

3.6.2 Width. Unless otherwise specified on the applicable drawings, contract, or order, the width of the bars shall be within the limits given in table II.

3.6.3 Length. Bars shall be furnished in lengths of at least 6 feet but not exceeding 14 feet.

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Table II

WIDTH TOLERANCE

Width (inch)	Permissible tolerance (inch)	
	Plus	Minus
Up to 1.00	0.008	0.005
Over 1.00 to 2 50 incl.	0.010	0.005
Over 2.50	0 015	0.010

3.6.4 Concavity The concavity at any point on one side of a bar shall not exceed 0.001 inch.

*3.6.5 Convexity The total convexity of both sides of a bar at any point shall not exceed 0.001 inch.

3.7 Workmanship. The bars shall be uniform in quality and temper and shall be clean and smooth, free from foreign material, pipes, slivers, laps, cracks, seams, scale, burrs, damaged edges, and any other defects which might render them unfit for the usage intended.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

*4.2 Lot. Unless otherwise specified, a lot shall consist of 10,000 pounds or fraction thereof of material from the same heat and manufactured by the same process and shall be of the same form, size, and temper, submitted for inspection at one time.

*4.3 Sampling. Unless otherwise specified, and when applicable, sampling plans and procedures shall be in accordance with the provisions of MIL-STD-105.

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4.4 Sampling for chemical analysis. The samples selected in accordance with 4.3 shall be analyzed in accordance with methods 111 or 112 of FED-STD-151. A single analysis of a composite sample may be made. In case of dispute, referee analysis shall be by method 111. If another method of analysis is desired, it shall be as agreed upon by the contractor and the procuring activity.

*4.5 Inspection. All material shall be inspected for surface defects and dimensions to assure compliance with dimensional requirements.

*4.6 Resistivity test. A resistivity test sample shall be selected in accordance with MIL-STD-105, acceptable quality level (AQL) of 1.0 percent. The resistivity test shall be conducted in accordance with ASTM B 193

*4.7 Hardness tests. Hardness tests shall be conducted in accordance with ASTM E 18.

*4.8 Acceptance and rejection. Each sample selected shall be inspected to meet the requirements of this specification. In the event the lot is rejected, it shall be subject to the acceptance and rejection provisions of MIL-STD-105.

5. PREPARATION FOR DELIVERY

5.1 Packing.

5.1.1 All material shall be separated by size when packed for shipment.

5.1.2 Unless otherwise specified, bars shall be bundled or packaged for shipment in such a manner as to protect the bars from any damage during handling or shipment.

5.1.3 Bundles or packages shall be so prepared as to insure acceptance by common or other carriers for safe transportation to the point of delivery at the lowest rate.

5.2 Labeling and marking. Unless otherwise specified, labeling and marking of all shipping containers shall be in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended use. Bars furnished under this specification are intended for use in commutators for torpedo motors.

*6.2 Ordering data. Unless otherwise specified, procurement documents should specify title, number, and date of this specification and exceptions to this specification and applicable documents. Dimensions of bars shall be specified in the contract or order, or on the applicable drawings referenced therein.

*6.3 Supersession data. This specification includes the requirements of MIL-B-19231 (NOrd) dated 30 December 1955, issued in lieu of NAVORD OS 3292, and Amendment 1, dated 22 May 1958. When drawings etc. refer to NAVORD OS 3292, this specification applies.

*6.4 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian.
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Preparing activity:
Navy - OS
(Project No. 9530-N010)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R5

INSTRUCTIONS This sheet is to be filled out by personnel either Government or contractor involved in the use of the specification in procurement of products for ultimate use by the Department of Defense This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost Comments and the return of this form will be appreciated Fold on lines on reverse side, staple in corner and send to preparing activity Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements

SPECIFICATION

MIL-B-19231A (OS), Bars, Copper, Silver-Bearing (For Commutators)

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes" in what way?)*

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification If there are additional papers attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED

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